JCTC Journal of Chemical Theory and Computation

Article

Cite This: J. Chem. Theory Comput. 2018, 14, 6175−6182

pubs.acs.org/JCTC

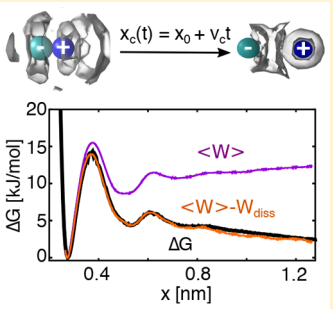
Targeted Molecular Dynamics Calculations of Free Energy Profiles Using a Nonequilibrium Friction Correction

Steffen Wolf* and Gerhard Stock*

Biomolecular Dynamics, Institute of Physics, Albert Ludwigs University, 79104 Freiburg, Germany

Supporting Information

ABSTRACT: As standard unbiased molecular dynamics (MD) simulations become impractical for sampling rare events, "targeted MD" employs a moving distance constraint to enforce rare transitions along some reaction coordinate x. To calculate free energy profiles from these nonequilibrium simulations via $\Delta G(x) = W(x) - W_{\mathrm{diss}}(x)$, apart from the (readily obtained) work $W(x)$ performed on the system, the dissipated work $W_{\mathrm{diss}}(x)$ is also required. By employing a second-order cumulant expansion of Jarzynski's equality combined with an analysis within Langevin theory, the dissipated work can be expressed via a nonequilibrium friction coefficient $\Gamma_{\mathrm{NEQ}}(x)$ that may be calculated on-the-fly from constraint force fluctuations. Adopting the ion dissociation of NaCl in water as a test system, this friction correction is shown to result in accurate free energy profiles, even for a modest number of simulations and at high constraint velocities. As a bonus, the analysis of $\Gamma_{\mathrm{NEQ}}(x)$ may yield valuable insight into the microscopic mechanism of friction.



Downloaded via MASARYK UNIV on November 7, 2024 at 00:09:46 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

INTRODUCTION

Let us consider a classical complex system, e.g., a protein in aqueous solution at constant pressure, that is in contact with a heat reservoir at temperature T. The system may exist in two equilibrium states A and B, and we are interested in the free energy difference $\Delta G = G_B - G_A$ of these conformational states.[1,2] To this end, one can perform molecular dynamics (MD) simulations of the system and monitor its time evolution along some reaction coordinate x that accounts for transitions between states A and B at positions x_A and x_B. By calculating the resulting probability distribution P along x, we obtain the free energy profile

$$\Delta G(x) = -k_{\mathrm{B}}T\ln[P(x)/P(x_A)] \tag{1}$$

from which we derive $\Delta G \equiv \Delta G(x_B)$. For simplicity, we restrict ourselves to a one-dimensional reaction coordinate x in the remainder of this paper.

Standard unbiased MD simulations become impractical if the states are separated by high energy barriers such that transitions between them occur only rarely. To this end, a number of enhanced sampling techniques[1−11] have been proposed, including, e.g., replica-exchange MD,[5] metadynamics,[6] and adaptive biasing force sampling.[7] To enforce rare transitions, in particular, one may employ some external force to pull the molecule from x_A to x_B. This is done, e.g., in simulations using moving harmonic restraints[8,9] along x such as steered MD[10,11] or in targeted MD simulations,[12−14] which employ moving distance constraints. As direct output, these simulations yield the work performed on the system by external pulling, which by the laws of thermodynamics is larger that the free energy difference,

$$W \geq \Delta G \tag{2}$$

where W denotes the work average. Here, the equal sign corresponds to an infinitely slow reversible process, while in a finite-time irreversible process, the difference $W - \Delta G = W_{\mathrm{diss}}$ accounts for the dissipated work associated with the increase of entropy.

To approximate a reversible "slow growth" process by pulling simulations, a thermodynamic integration scheme may be employed, which calculates the free energy difference via the potential of mean force,[15]

$$\Delta G_{\mathrm{TI}} = \int_{x_A}^{x_B} \frac{\mathrm{d}G}{\mathrm{d}x}\mathrm{d}x = \int_{x_A}^{x_B} \langle f_{\mathrm{c}}\rangle_x \mathrm{d}x \approx \sum_i \langle f_{\mathrm{c}}\rangle_{\mathrm{eq},x_i}\Delta x \tag{3}$$

where $\langle f_{\mathrm{c}}\rangle_{\mathrm{eq},x}$ represents an equilibrium average of the pulling force f_{c} at point x. While representing a straightforward and well-established approach to compute ΔG, thermodynamic integration is in practice quite demanding, because it typically requires numerous (i.e., for each x_i) and relatively long MD simulations to converge to equilibrium.

On the other hand, several ideas have been proposed to calculate $\Delta G(x)$ directly from the pulling trajectories. In particular, Jarzynski[16] showed for a general nonequilibrium process that the free energy difference can be directly calculated from the work W via

$$\Delta G = -k_{\mathrm{B}}T\ln\langle \mathrm{e}^{-W/k_{\mathrm{B}}T}\rangle \tag{4}$$

Here $\langle\ \rangle$ denotes an ensemble average over many independent nonequilibrium simulations with initial condition sampled

Received: August 10, 2018
Published: November 8, 2018

© 2018 American Chemical Society

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

from the underlying canonical ensemble. As the individual simulations in principle can be of arbitrary short duration, a "fast growth" implementation[17] of eq 4 has been suggested that may be advantageous compared to the slow growth method based on thermodynamic integration.

In practice, however, the evaluation of Jarzynski's equality requires a very large number of nonequilibrium trajectories because the exponential average $\langle e^{-W/k_BT}\rangle$ is dominated by small values of the work which occur only rarely.[11,18] Alternatively, a cumulant expansion, $\ln\langle e^x\rangle = \langle x\rangle + (\langle x^2\rangle - \langle x\rangle^2)/2 + ...$, of eq 4 has been considered.[11,17] Truncating in second order, this yields with $\delta W = W - \langle W\rangle$

$$\Delta G = \langle W\rangle - \frac{1}{2k_BT}\langle \delta W^2\rangle \tag{5}$$

which is an exact expression if the work distribution is Gaussian. It states that the dissipated work $W_{diss} = \langle W\rangle - \Delta G$ is given in terms of the variance of the work fluctuations. While the cumulant approximation in eq 5 converges more rapidly than Jarzynski's expression in eq 4, it is *a priori* not clear if the lower tail of the work distribution (which contains the main contribution to the free energy) is well represented by a Gaussian.

In this article, we propose an alternative approach to calculate the free energy difference $\Delta G(x)$ directly from nonequilibrium trajectories, which are generated in constraint-based targeted MD (TMD) simulations.[12−14] To this end, we model the molecular system by a Langevin equation and describe the TMD pulling by an external force. Assuming a constant constraint pulling velocity v_c, our analysis leads to the simple relation

$$\Delta G = \langle W\rangle - v_c \int_{x_A}^{x_B} \Gamma(x)\mathrm{d}x \tag{6}$$

where $\Gamma(x)$ represents the position-dependent friction coefficient of the system, which can be calculated using standard methods.[19−23] Eq 6 provides a link of the dissipated work W_{diss} with the friction $\Gamma(x)$ and thus offers a microscopic picture of the system−bath coupling. Moreover, we show that the cumulant approximation (eq 5) can also be recast in the form of eq 6, leading to a new expression for a nonequilibrium friction coefficient.

Adopting the ion pair dissociation of sodium chloride in water as a simple yet nontrivial model system,[24−26] we illustrate and test this relation by comparing its outcome to results of unbiased MD and thermodynamic integration calculations, which serve as a reference. We demonstrate that eq 6 provides an easy-to-calculate and accurate estimate of $\Delta G(x)$, as long as the pulling speed v_c is slow compared to the time scale of the bath fluctuations in the Langevin model representing all other motions than along x of the system. Calculation of $\Gamma(x)$ directly from nonequilibrium simulations accounts for nonlinear velocity-dependent effects, resulting in an excellent friction correction over a broad range of pulling velocities.

THEORY

To derive eq 6, in the following, we analyze the unbiased dynamics of molecular system by a nonoverdamped Langevin equation and subsequently describe the effect of the pulling in TMD simulations by an external force in the Langevin equation. Accounting for rare events, the motion along the reaction coordinate x is supposed to be slow compared to the remaining degrees of freedom of the molecule. Exploiting this time scale separation between the slow motion of the "system" and the fast fluctuations of the "bath", the unbiased motion along x can be described by a memory-free Langevin equation,[15,27]

$$m\ddot{x}(t) = f(x) - \Gamma(x)\dot{x} + K(x)\xi(t) \tag{7}$$

which contains a Newtonian force term $f = -\mathrm{d}G/\mathrm{d}x$ with $G(x)$ being the potential of mean force, a Stokes' term $-\Gamma\dot{x}$ with the friction field $\Gamma(x)$, as well as a stochastic force $K\xi(t)$ with amplitude $K(x)$ and "noise" $\xi(t)$, that usually is assumed to be of zero mean, $\langle\xi\rangle = 0$, delta-correlated, $\langle\xi(t)\xi(t')\rangle = \delta(t - t')$, and Gaussian distributed.

As detailed in the Methods section, TMD simulations enforce rare transitions between conformational states A and B by applying a constraint force f_c along the reaction coordinate x. While external driving forces in general need to be taken into account in the derivation of the Langevin equation,[28] we assume for the moment that the constraint force does not change the Langevin fields f, Γ, and K but can be simply included as an additive term in eq 7. The conditions for this assumption are derived below. This yields the nonequilibrium Langevin equation

$$m\ddot{x}(t) = -\frac{\mathrm{d}G}{\mathrm{d}x} - \Gamma(x)\dot{x} + K(x)\xi(t) + f_c(t) \tag{8}$$

In the particular case of a TMD simulation, the constraint force f_c effects a constant velocity of the system,

$$\dot{x} = v_c \tag{9}$$

which results in a vanishing total force, $m\ddot{x} = 0$. To achieve this cancellation, the constraint force f_c needs to (i) counterbalance the effect of the stochastic force and (ii) keep the velocity constant against the drag of the friction force and the acceleration due to the free energy gradient. To study if these requirements are consistent with the simple form assumed for the nonequilibrium Langevin eq (eq 8), we first focus on the second point.

Assuming that the pulling speed v_c is slow compared to the time scale of the bath fluctuations, the latter effect of f_c can be considered as a slow adiabatic change,[29−31] which leaves the system virtually at equilibrium at all times. In the limiting case of $v_c = 0$, in particular, this corresponds to a position constraint (i.e., x = const.) of the system. This constraint can be realized via a bilinear coupling of the system coordinate x to a high-frequency harmonic restraint mode (so-called "stiff spring" approximation),[11] which indeed can be shown to result in a simple additive force term in the Langevin equation.[32] While harmonic restraints and holonomic constraints represent two different mathematical concepts, their effect of fixing the motion along x becomes equivalent in the limit of stiff springs. To derive the unbiased potential of mean force from the potential of mean constraint force, one can apply a Fixman determinant correction which is zero for distance-type reaction coordinates.[14,33]

The situation is different for the other mentioned effect of f_c (cancellation of the stochastic force) because the time scales of this part of the external force and the bath fluctuations are necessarily the same. However, the derivation of eq 6 does not require that the constraint force f_c satisfies eq 9 for an individual stochastic trajectory. Rather, it is sufficient, if we assume that f_c keeps the velocity constant *on average*. To see

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

this, we eliminate the effects of the stochastic force and the details of the initial preparation of the TMD simulation by performing an ensemble average of eq 8 over many TMD runs, which yields

$$\frac{\mathrm{d}G}{\mathrm{d}x} = -\Gamma(x)v_c + \langle f_c(x)\rangle \tag{10}$$

where we have used that $\langle\xi\rangle = 0$, $\langle\dot{x}\rangle = v_c$, and $\langle x(t)\rangle = x_A + v_c t$. Note that eq 10 does not require that f_c counterbalances the effect of the stochastic force in eq8 but only needs to effect that $\langle\dot{x}\rangle = v_c$. Hence, we showed that the nonequilibrium Langevin eq 8 is valid in case the constraint velocity v_c is slower than typical bath motions.

Having justified our ansatz in eq 8, we go on and integrate on both sides of eq 10 from initial position x_A to final position x. This yields the desired relation

$$\begin{aligned}\Delta G(x) &= \int_{x_A}^{x} \langle f_c(x')\rangle \mathrm{d}x' - v_c \int_{x_A}^{x} \Gamma(x')\mathrm{d}x' \\ &= \langle W(x)\rangle - W_{\mathrm{diss}}(x)\end{aligned} \tag{11}$$

where $\langle W(x)\rangle$ represents the averaged external work performed on the system.

Several comments are in order. First, we have shown that the dissipated work arising in a TMD simulation can be written as a Stokes friction term, that can be readily calculated using[19−23]

$$\Gamma(x) = \frac{1}{k_B T}\int_0^\infty \langle \delta f_c(t)\delta f_c(0)\rangle_{\mathrm{eq},x}\mathrm{d}t \tag{12}$$

where the autocorrelation function of the force $\delta f_c = f_c - \langle f_c\rangle$ is obtained from an equilibrium MD simulation where the position of the reaction coordinate is constrained to the value x. Second, we wish to mention that the above derivation is readily extended to the case of a generalized Langevin equation, which exhibits a friction term $-\int_0^t \mathrm{d}t'\,\zeta(t-t')\,\dot{x}(t')$ with memory kernel $\zeta(t-t')$ instead of the memory-free friction term $-\Gamma\dot{x}$ in eq 7. Assuming constant velocity, we obtain $-v_c\int_0^t \mathrm{d}t'\,\zeta(t')$ which in effect corresponds to memory-free friction.

It is instructive to relate the above derived expression for W_{diss} to the corresponding result (eq 5) from second-order cumulant expansion of Jarzynski's equality. This yields

$$\begin{aligned}W_{\mathrm{diss}} &= \frac{1}{2k_B T}\left\langle\left(\int_{x_A}^{x_B}\delta f_c(x)\mathrm{d}x\right)^2\right\rangle \\ &= \frac{1}{k_B T}\int_{x_A}^{x_B}\mathrm{d}x_2\int_{x_A}^{x_2}\mathrm{d}x_1\langle\delta f_c(x_2)\delta f_c(x_1)\rangle\end{aligned} \tag{13}$$

where the missing factor 2 in the second line is caused by changing the upper limit of the first integral from x_A to x_2. Comparison to eq 11 suggests that within second-order cumulant approximation the friction term is given by

$$\begin{aligned}\Gamma_{\mathrm{NEQ}}(x) &= \frac{1}{k_B T v_c}\int_{x_A}^{x}\langle\delta f_c(x)\delta f_c(x_1)\rangle\mathrm{d}x_1 \\ &= \frac{1}{k_B T}\int_0^{t(x)}\langle\delta f_c(t)\delta f_c(t-\tau)\rangle\mathrm{d}\tau\end{aligned} \tag{14}$$

where in the second line we changed variables using $x_1 = x_A + v_c t'$ and $\tau = t - t'$. That is, the friction arising in a TMD simulation is proportional to the time integration of an autocorrelation function of the constraint force. Together with eq6, the nonequilibrium friction coefficient in eq14 represents the central theoretical result of this work. The subscript of $\Gamma_{\mathrm{NEQ}}(x)$ reminds us that the autocorrelation function is obtained via an ensemble average over many nonequilibrium trajectories, as opposed to a time average in equilibrium friction calculations, and therefore is distinctly different from $\Gamma(x)$ from eq 12 despite their similar structure. Equilibrium (eq 12) and nonequilibrium (eq 14) expressions for the friction coefficient are thus based on different physical assumptions and may lead to different results for finite velocities v_c. We note in passing that it has been shown recently[34] that constraints may affect the absolute value of the calculated friction coefficient.

METHODS

MD Details. All simulations of NaCl in water were carried out using Gromacs 4.6.7. (ref 35). The simulation system consisted of a cubic box of 3 nm side length filled with 895 TIP3P water molecules[36] as well as one sodium and one chloride ion, which were described by the Amber99 ion parameters.[37] We used a 1 fs time step and constrained bonds and angles of the water molecules by the SETTLE algorithm.[38] Electrostatics were described by the particle mesh Ewald (PME) method.[39] Cutoffs were set to 1 nm for van der Waals interactions and PME real space. The simulations were carried out in a NPT ensemble with the temperature set to 293.15 K and the pressure to 1 bar. Temperature control was achieved by the Bussi velocity rescaling thermostat[40] (coupling time τ_T = 0.2 ps), and pressure control via the Parrinello−Rahman barostat[41] (isotropic pressure coupling, coupling time τ_P = 0.5 ps, compressibility of 4.5×10^{-5} bar^{-1}). After a short equilibration phase of 1 ns, the unbiased equilibrium production simulation was run for a total trajectory length of 200 ns. Unbiased free energies were then calculated according to eq 1, where $P(x)$ was calculated by histogramming the distance data between 0.245 nm and 2.615 ns with a time resolution of 0.1 ps and a spacial resolution of 5 pm. Visualization of molecular data was performed with vmd.[42]

Targeted Molecular Dynamics. TMD simulations[12−14] pull the system along a reaction coordinate x with constant velocity v_c. That is, we drive the system from $x(0) = x_A$ to $x(t_B) = x_B$ by constraining coordinate $x(t)$ to $x_c(t) = x_A + v_c t$ via the constraint function

$$\Phi(x(t)) = x(t) - x_c(t) = 0 \tag{15}$$

The constraint is realized via the force

$$f_c = \lambda\frac{\mathrm{d}\Phi(x(t))}{\mathrm{d}x} = \lambda \tag{16}$$

where λ is a Lagrange parameter, which is determined at each time step of the TMD simulations in a self-consistent manner.[12−14]

TMD simulations of NaCl in water were performed employing the PULL code as implemented in GROMACS 4.6.7 (ref 35) using the "constraint" mode based on the SHAKE algorithm.[43] The pulling coordinate was chosen as the distance between the Na^+ and Cl^- ions. All simulations started from the equilibrated system structure after the initial 1 ns NPT run, using the same thermostat and barostat scheme. Following a short initial TMD run to adjust the interion distance to 0.265 nm, the system was again equilibrated for

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

1 ns. Constant velocity simulations were then performed with v_c = 1, 5, 10, 25, 50, 100, 125, and 250 nm/ns, respectively. The simulation length was adjusted such that the ions reached the maximal interion distance of 1.265 nm within the simulated time. Constraint forces were written out each time step. Thermodynamic integration averages (eq 3) were calculated from 10 ns long TMD runs at fixed position x_i (i.e., for v_c = 0), using only the last 2.5 ns to evaluate $\langle f_c \rangle_{x_i}$. Typically, N = 21 equidistant points were used to approximate the integral.

Calculation of Friction Coefficients. The above-mentioned constant distance simulations were also used to calculate the equilibrium force autocorrelation function in eq 12. Moreover, water distributions around NaCl ions were calculated as occupancy probabilities (so-called water densities).[44] The isosurface encloses a volume with a water oxygen occupancy probability of >0.57 Å^{-3}.

To calculate nonequilibrium observables such as work $\langle W(x) \rangle$ and nonequilibrium friction $\Gamma_{NEQ}(x)$, an ensemble average over N trajectories is required. While in the case of $\langle W(x) \rangle$ already N = 30 trajectories suffice (Figure S1), the convergence of $\Delta G(x)$ calculated via a $\Gamma_{NEQ}(x)$-based dissipation correction depends on the pulling velocity (N = 30 for 1 nm/ns vs N = 1000 for 50 nm/ns, see Figures S2 and S3). In all calculations reported below, N = 1000 trajectories are used. For all investigated velocities, both δf_c and δW follow a Gaussian distribution (see Figure S4). While the nonequilibrium force autocorrelation function and the associated friction $\Gamma_{NEQ}(x)$ are found to fluctuate considerably, the eventually desired dissipated work $W_{diss}(x)$ (obtained via integration of $\Gamma_{NEQ}(x)$) is smooth and well-behaved (Figure S5).

RESULTS AND DISCUSSION

Unbiased MD and Thermodynamic Integration. To obtain a first overview of the NaCl/water system, Figure 1a shows the free energy profile $\Delta G(x)$ along the NaCl ion pair distance, calculated from 200 ns unbiased MD simulation. $\Delta G(x)$ is seen to exhibit a global minimum at x_0 = 0.275 nm which corresponds to the bound (associated) state, and a global maximum at x_1 = 0.375 nm corresponding to the dissociation/association transition state. Secondary local minima and maxima can be found at x_3 = 0.54 nm and x_4 = 0.615 nm, respectively, reflecting the formation of a second hydration shell between the ion pair.[26] The dynamics of ion dissociation and association can be described by the time series and autocorrelation function of the ion distance x (Figure S6) as well as by the first passage time distributions (Figure 1b) of these processes. Although the energy barrier is somewhat higher for dissociation than for association, the mean first passage time of 150 ps for the former process is significantly shorter than the 2.6 ns obtained for the latter. This finding indicates that apart from the interion distance, some kind of solvent coordinate needs to be invoked in order to microscopically explain the mechanism of the ion pair association.[26]

Using the results from the unbiased MD simulation as a reference, we next consider various ways to calculate the free energy from nonequilibrium TMD simulations (see Methods for details). Beginning with the most common method, Figure 1c shows results from thermodynamic integration (eq 3). We note that this calculation is particularly straightforward for TMD simulations because the (negative) mean potential of the constraint force corresponds directly to the free energy profile.[14] Nonetheless, finding a suitable spacing of the "windows" (i.e., equilibration points along x) is both crucial and not straightforward. For example, by naively using 21 equidistant windows along the coordinate range of interest, the thermodynamic integration results of $\Delta G(x)$ show significant deviations from the reference. This is because equidistant points do not necessarily capture the important extremal points of $\Delta G(x)$. Only when we include these points (which means "cheating" because they are not known *a priori*), the thermodynamic integration results are equivalent to the unbiased reference.

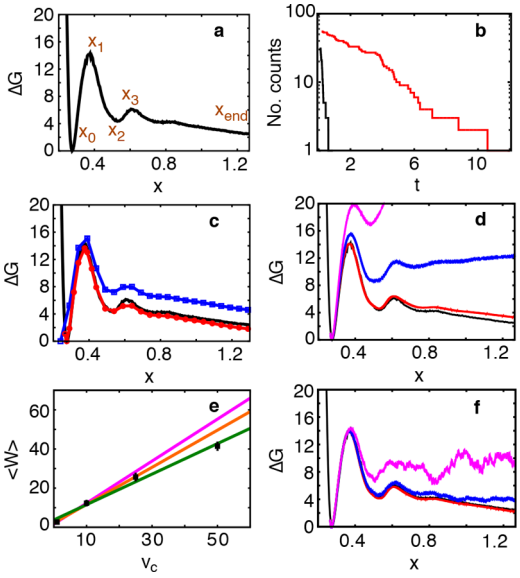


Figure 1. (a) Free energy profile $\Delta G(x)$ (in kJ/mol) describing the NaCl ion pair in water as a function of the interion distance x (in nm), as calculated from unbiased MD simulations. Characteristic distances x_0 to x_{end} used for further analysis are indicated. (b) First passage time distribution (in ns) for NaCl ion pair dissociation (black) and association (red), yielding mean first passage times k_{diss} = 1/150 ps^{-1} and k_{ass} = 1/2600 ps-1, respectively. (c) Comparison of $\Delta G(x)$ obtained from unbiased MD (black) and thermodynamic integration using equidistant (blue) and adjusted (red) equilibration points (see text). (d) Work profiles $\langle W(x) \rangle$ (in kJ/mol) obtained for v_c = 1 nm/ns (red), v_c = 10 nm/ns (blue) and v_c = 50 nm/ns (magenta), compared to the unbiased free energy (black). (e) Mean work $\langle W \rangle$ at $x = x_{end}$ plotted as a function of v_c (in nm/ns). Fits using the first two points are in magenta, first three points in orange, and all points in green. (f) $\Delta G(x)$ calculated according to Jarzynski's identity (eq 4) using the same color code as in panel d.

TMD Simulations. The findings above clearly indicate the advantage of slow growth methods, which sample the conformational space along x in a continuous way and therefore avoid errors due to omitted extrema. With this end in mind, we now turn to the main topic of this work, that is, the direct nonequilibrium evaluation of the TMD pulling simulations. In a first rough approximation, we neglect the dissipated work $W_{diss}(x)$ and focus on the work $\langle W(x) \rangle$ performed by the pulling. Comparing this work to the correct free energy, Figure 1d reveals that $\langle W(x) \rangle$ only agrees with $\Delta G(x)$ at rather slow velocities (e.g., v_c = 1 nm/ns), while it considerably overestimates the free energy already at v_c = 10 nm/ns.

Nonetheless, there are ways to calculate $\Delta G(x)$ solely from $\langle W(x) \rangle$, which do not simply neglect the dissipated work. For

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

example, by assuming that $W_{diss}(x) \propto v_c$ (cf. eq 11), we may plot the work as a function of v_c and estimate ΔG from linear regression to $v_c = 0$. Fixing the position to $x = x_0$ and $x' = x_{end}$, Figure 1e shows that the linearity in v_c is maintained up to $v_c \approx$ 25 nm/ns. Table 1 and Figure S7 reveal that this simple and computationally quite inexpensive method already yields very good agreement (within $\lesssim$1 kJ/mol) with the unbiased free energy.

Table 1. Estimates of the Free Energy $\Delta G = \Delta G(x_{end})$ (kJ/mol) as Obtained for Various Nonequilibrium Methods and Pulling Velocities v_c (nm/ns)[a]

v_c	$\langle W \rangle$	ΔG_{LR}	ΔG_{Cr}	ΔG_{Γ}	ΔG_{Jz}	$\Delta G_{\Gamma_{NEQ}}$
1	2.7		2.4	2.5	2.2	2.2
10	12.3	1.6	1.9	0.9	3.7	2.2
25	25.6	2.1	3.8	−6.2	6.6	2.2
50	41.4	3.7	2.1	−17.6	8.7	3.9

[a]Index LR stands for linear regression (cf. Figure 1e), Cr for Crooks' theorem, Jz for Jarzynski's equality (eq 4), and Γ and Γ_{NEQ} for equilibrium and nonequilibrium friction calculations, respectively; see text. The reference value from unbiased MD is ΔG = 2.5 kJ/mol.

Alternatively, we may invoke Crooks' theorem[45] which allows us to estimate ΔG from the ratio of the probability distributions of the work performed in forward and backward direction (i.e., enforced dissociation and association, respectively). As shown in the SI (Table S1) and in Table 1, the procedure leads to reasonable values for ΔG at velocities $v_c \lesssim$ 50 nm/ns.

To obtain the full free energy profile $\Delta G(x) = W(x) - \langle W_{diss}(x) \rangle$, we employed Jarzynski's identity eq 4. Figure 1f compares the resulting free energy profiles $\Delta G(x)$ to reference results from unbiased MD. We obtain good results for $v_c \lesssim 10$ nm/ns, which represents a clear improvement over the results of thermodynamic integration using the same window positions (Figure 1c). For higher velocities, the dissipated work is underestimated. This represents a common problem of the direct application of eq 4, which relies heavily on the correct sampling of the low-W tail of the underlying probability density.[46]

As an alternative approach, we next explicitly calculate the dissipated work $W_{diss}(x) = v_c \int_{x_A}^{x} \Gamma(x')\, dx'$. We first use the equilibrium expression (eq 12) for the friction coefficient $\Gamma(x)$, which should yield valid estimates of $\Delta G(x)$ as long as the pulling velocity v_c is slow compared to the time scale of the bath fluctuations. Considering velocities v_c from 10 to 100 nm/ns, Figure 2 displays the resulting free energy profiles $\Delta G(x)$ together with the results from unbiased MD. Similarly to the corrections based on eq 4, we obtain good results for $v_c \lesssim 10$ nm/ns. For larger velocities, the assumed time scale separation between pulling and bath fluctuations appears to break down, which affects an overestimation of the friction.

On the other hand, we may employ the nonequilibrium friction coefficient $\Gamma_{NEQ}(x)$ (eq 14) that should be appropriate as long as the distribution of the work is Gaussian. A major advantage herein is that, by predefining the shape of the probability density, all computed samples contribute to the variance estimation, thus achieving an improved convergence in comparison to eq 4. In fact, the nonequilibrium friction is seen to achieve excellent results even at high velocities of 50 nm/ns. For this velocity, the friction correction correctly

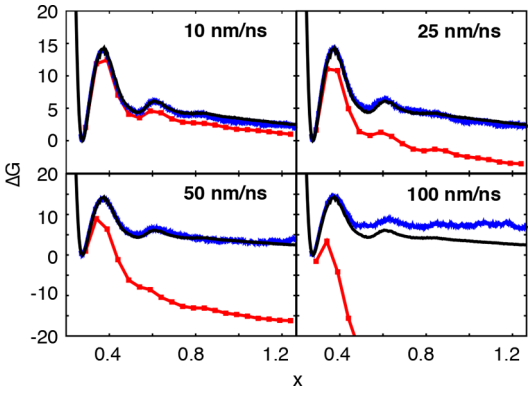


Figure 2. Free energy profiles $\Delta G(x)$ (in kJ/mol) along x (in nm) obtained for pulling velocities v_c as indicated. Compared are results using eq 11 with equilibrium (red) and nonequilibrium (blue) friction coefficients to reference results from unbiased MD (black).

results in $\Delta G(x_{end}) \approx 1\ k_B T$, despite an overlaying $W_{diss} \approx 15\ k_B T$. At velocities $v_c \geq 100$ nm/ns, the predicted free energy profile diverges from the reference profile. This divergence in part originates from an insufficient action of the thermostat at the given coupling time, causing a perturbation of the bath velocity distribution at high pulling velocities. Moreover, the underlying assumption of a Gaussian work distribution is expected to break down for high velocities. To test this assumption, we performed a d'Agostino-Pearson omnibus test[47] which gives a p-value between 0 and 1, the latter corresponding to a perfect Gaussian. Interestingly, we find already at low velocities (e.g., 10 nm/ns, for which the nonequilibrium friction correction yields excellent results) relatively small p-values (e.g., 0.55), see Table S1. For v_c = 100 nm/ns, the p-value drops to 0.16, which indicates a breakdown of the Gaussian assumption.

Molecular Origin of Internal Friction. The insight that friction depends on position and velocity and may significantly determine the physics of the NaCl/water system brings up the question on its molecular origin.[48−52] To gain a microscopic understanding, we first focus on the equilibrium case (eq 12) which is determined by the force autocorrelation function.[19−23] As shown in Figure S8, this function is well approximated by a biexponential fit, $\langle \delta f_c(t) \delta f_c(0) \rangle_x = \langle \delta f_{c_1}^2 \rangle\, e^{-t/\tau_1} + \langle \delta f_{c_2}^2 \rangle\, e^{-t/\tau_2}$, with a fast component $\tau_1 \approx 18$ fs and a slow component $\tau_2 \approx 1$ ps. Time integration yields via eq 12 the equilibrium friction coefficient

$$\Gamma(x) = \frac{1}{k_B T}(\langle \delta f_{c_1}^2 \rangle \tau_1 + \langle \delta f_{c_2}^2 \rangle \tau_2) = \Gamma_1(x) + \Gamma_2(x) \quad (17)$$

Showing the two contributions $\Gamma_1(x)$ and $\Gamma_2(x)$, Figure 3a reveals that the fast component is constant almost over the full range of x. Moreover, the 18 fs decay time is found to be in perfect agreement with the decay of the autocorrelation function of the ion−water contacts (i.e., of the distance of Na^+ or Cl^- to the closest water molecule, see Figure S9). These findings indicate that $\Gamma_1(x)$ is caused by thermal collisions between the ions and their surrounding water molecules. This also explains that $\Gamma_1(x)$ decreases for $x \lesssim 0.4$ nm, as at such distances no water is present between the ions, which reduces the number of collisions. The femtosecond time scale of τ_1 highlights the necessity to use constraint (over harmonic

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

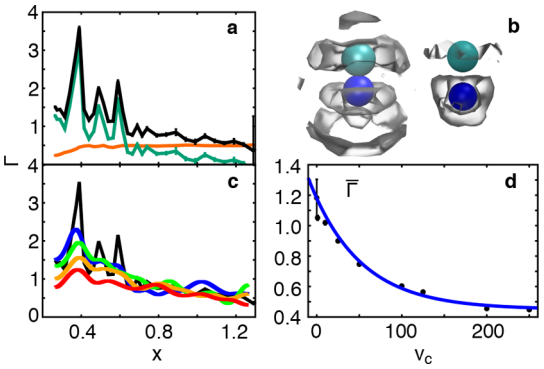


Figure 3. Molecular origin of the friction of the NaCl/water system. (a) Equilibrium friction $\Gamma(x)$ (black) (in kJ ns/mol nm^2) together with its fast (orange) and slow (green) components $\Gamma_1(x)$ and $\Gamma_2(x)$, respectively. (b) Density plots of water molecules surrounding the Na (blue) and Cl (cyan) ions, taken at x = 0.275 nm (left) and 0.39 nm (right). (c) Comparison of equilibrium friction $\Gamma(x)$ (black) and nonequilibrium friction $\Gamma_{NEQ}(x)$. The latter is shown for velocities v_c = 1 (blue), 10 (green), 25 (orange) and 50 (red) nm/ns. (d) Velocity dependence (in nm/ns) of the averaged friction $\bar{\Gamma}_2$ together with the exponential fit of eq 18 (red line).

restraint) pulling, as it allows for the correct sampling of dynamics effecting $\Gamma_1(x)$.

The slow component of the friction, $\Gamma_2(x)$, on the other hand, shows a significant position dependence. It exhibits three maxima at 0.39, 0.49, and 0.59 nm and vanishes for large x. The maxima at 0.39 and 0.59 nm correlate well with the maxima found for $\Delta G(x)$, which can be associated with the formation of solvation shells. The maximum at 0.49 nm coincides with a shoulder, i.e., a shallow maximum in the $\Delta G(x)$ profile (cf. Figure 1). Figure 3b shows density plots of the water molecules surrounding the ions taken at the first minimum and the first maximum of the friction. (Density plots of the other extrema are discussed in Figure S10.) We find that the minima of the friction are associated with a highly ordered hydrogen bonding network, e.g., reflected by three tori of water density at x = 0.275. At the maxima of $\Gamma_2(x)$, this ordering vanishes due to the increased flexibility of the system. At large ion distance, $\Gamma_2(x)$ decreases because the ions do no longer possess a common hydration shell. Moreover, we find that the autocorrelation decay time of ≈1 ps is in good agreement with the time scale of collective hydrogen bond vibrations in water molecules surrounding host molecules,[53] and is also in the order of the transfer time between x_0 and x_2 in unbiased simulations. Taken together, we conclude that the position dependence of $\Gamma_2(x)$ originates from collisions with water molecules vibrating in collective modes caused by ordered hydration shells surrounding the ions. In this way, $\Gamma_2(x)$ provides valuable insight on the hydration dynamics of the system, which reaches well beyond the information given by $\Delta G(x)$ alone.

It is interesting to compare the above results for the equilibrium friction $\Gamma(x)$ to the results obtained from eq 14 for the nonequilibrium friction $\Gamma_{NEQ}(x)$. Unlike for the equilibrium case, $\Gamma_{NEQ}(x)$ depends explicitly on the pulling velocity v_c and should approach $\Gamma(x)$ in the limit of small v_c. Figure 3c confirms these expectations and reveals that only for v_c = 1 nm/ns all three maxima of $\Gamma(x)$ are resolved by $\Gamma_{NEQ}(x)$. Reflecting the ordered hydration shells of the ions, these sharp features are smoothened at higher velocities due to the short persistence time of the ions in these solvation structures. For x ≳ 0.7 nm, equilibrium and all nonequilibrium friction coefficients are roughly the same.

Analysis of the nonequilibrium force autocorrelation function (Figure S5) shows that the nonequilibrium friction, too, can be characterized by fast and slow components Γ_1 and Γ_2 with decay times of 18 fs and 1 ps, respectively. The fast component again produces the (roughly) constant background of the friction profiles. As the pulling occurs on a much longer time scale than 18 fs, this component does not depend on the pulling velocity. Hence, the intriguing velocity dependence of the nonequilibrium friction is exclusively caused by the slow component Γ_2, which accounts for the reordering of the hydration shells. Introducing the averaged friction coefficient $\bar{\Gamma} = (x_{end} - x_0)^{-1} \int_{x_0}^{x_{end}} dx \Gamma(x)$, Figure 3d shows the velocity dependence of the averaged slow component of the nonequilibrium friction, which decreases by about 60% for high velocities. This behavior is well-captured by the phenomenological ansatz

$$\bar{\Gamma}(v_c) = \bar{\Gamma}_1 + \bar{\Gamma}_2 e^{-v_c/v_{crit}} \tag{18}$$

where $\bar{\Gamma}_k$ are the averaged equilibrium components and $v_{crit} \approx$ 60 nm/ns represents the critical velocity, at which the pulling approaches the picosecond time scale of the reordering of the solvation shells, resulting in the breakdown of the system−bath time scale separation for the slow friction component. Consequentially, velocities $v_c \gtrsim v_{crit}$ result in the vanishing of Γ_2. The origin of the velocity dependence of Γ_{NEQ} (which seems to be in variance with the Markovian framework of Langevin eq 7) is its derivation from Jarzinsky's equality that warrants its validity even in the deep nonequilibrium regime.

CONCLUSIONS

Inspired by pioneering papers of Jarzynski and Crooks, several groups have invoked nonequilibrium methods to calculate equilibrium free energies.[18] Extending these efforts, in this work we combined TMD simulations and a Langevin-based analysis of these nonequilibrium simulations. We chose to use constraint-based TMD instead of harmonic restraint-based methods as it ensures a constant sampling along the reaction coordinate at integrator step size, allows for a straightforward calculation of the potential of mean force, and does not cause numerical problems at steep gradients. Adopting a Langevin formulation, we showed that the dissipated work $W_{diss}(x)$ can be expressed in term of a position-dependent friction coefficient $\Gamma(x)$. Employing second-order cumulant approximation of Jarzynski's equation, we derived a nonequilibrium friction coefficient Γ_{NEQ} which is valid as long as the distribution of the work is Gaussian.

Adopting the ion dissociation of NaCl in water as a well-established model system, we compared various methods to calculate free energy profiles $\Delta G(x)$ along the ion distance x, see Table 1. Considering first methods that only need to compute the performed work W, we proposed to calculate ΔG from linear regression of $W(v_c)$ to v_c = 0. Using pulling velocity $v_c \lesssim$ 25 nm/ns, this simple and inexpensive method was shown to yield very good agreement (within ≲1 kJ/mol) with the unbiased free energy. On the other hand, by explicitly calculating the dissipated work $W_{diss}(x)$ from a nonequilibrium approximation of the friction coefficient, we obtained excellent results for $\Delta G(x)$ for velocities up to 50 nm/ns. Comparing our own effort of standard thermodynamics integration calculations (here 21 equilibrium simulations of each 10 ns

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

length) to the effort of the nonequilibrium calculation at v_c = 50 nm/ns (100 nonequilibrium trajectories of each 20 ps length), this results in a speed-up of 2 orders of magnitude.

It is clear that our results for the relatively simple example of NaCl in water do not necessarily generalize to more complicated situations, e.g., when we pull a ligand out of a binding pocket. Popular methods to treat such complex systems are umbrella sampling[8] and steered MD[10,11] combined with WHAM[54] as well as metadynamics.[6] Here, we considered several nonequilibrium approaches based on Jarzynski's equality, combined with either direct evaluation, equilibrium-based friction, or cumulant expansion-based friction. The latter was found to outperform the convergence of direct evaluation significantly, although the basic sampling problem also applies, of course, to the force correlation functions in eqs 12 and 14. The price for the improved convergence is the presumption that the work follows a Gaussian distribution. In our NaCl/water test case, a clear time scale separation is given due to the fast dynamics of bulk water. Moreover, the system itself is isotropic; hence, the pathway of dissociation is well-defined. In more complicated systems this may not be the case, leading to dynamic correlations, e.g., between protein and ligand as well as the overlap of several $W_{diss}(x)$ profiles, which result in non-Gaussian distributions. As both problems originate from the presence of hidden reaction coordinates, the choice of an appropriate pulling coordinate remains challenging.[55]

With this efficient dissipation correction method at hand, we currently extend our studies to more complex molecular systems. We expect advantages for the study of transition path ensembles found in nonequilibrium simulations of protein functional dynamics.[56] Efficiency turns out to be crucial here because larger systems with many degrees of freedom may require extensive sampling (i.e., many nonequilibrium trajectories). The calculation of near-continuous $\Delta G(x)$ and $\Gamma(x)$ fields allows for coarse-graining in the form of Langevin simulations,[57] which can be used for the calculation of kinetic properties, e.g., association and dissociation kinetics of protein−ligand complexes.[58] Lastly, the calculation of velocity-dependent nonequilibrium friction factors may be beneficial for the analysis of friction in soft-matter systems in general and thus contribute to the understanding of microscopic mechanisms leading to protein-internal friction[48−52] or shear thinning or thickening fluids.[59]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.8b00835.

Convergence of work, friction, and free energy; Gaussianity of the work distribution; autocorrelation function for NaCl dissociation and association; linear regression of $\langle W_{v_c} \rangle$; application of Crooks' theorem; constraint force autocorrelation function; and water density profiles (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

*E-mail: steffen.wolf@physik.uni-freiburg.de.
*E-mail: stock@physik.uni-freiburg.de.

ORCID

Steffen Wolf: 0000-0003-1752-6175

Gerhard Stock: 0000-0002-3302-3044

Funding

This work was supported by the Deutsche Forschungsgemeinschaft (Sto 247/11) and the bwUniCluster and bwForCluster computing initiatives of the State of Baden-Württemberg.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank Sebastian Buchenberg, Matthias Ernst, Benjamin Lickert, Tanja Schilling, Jürgen Schlitter, and Florian Sittel for instructive and helpful discussions.

■ REFERENCES

(1) Chipot, C.; Pohorille, A. *Free Energy Calculations*; Springer: Berlin, 2007.
(2) Christ, C. D.; Mark, A. E.; van Gunsteren, W. F. Basic Ingredients of Free Energy Calculations: A Review. *J. Comput. Chem.* **2009**, *31*, 1569−1582.
(3) Fiorin, G.; Klein, M. L.; Henin, J. Using collective variables to drive molecular dynamics simulations. *Mol. Phys.* **2013**, *111*, 3345−3362.
(4) Tribello, G. A.; Bonomi, M.; Branduardi, D.; Camilloni, C.; Bussi, G. PLUMED 2: New feathers for an old bird. *Comput. Phys. Commun.* **2014**, *185*, 604−613.
(5) Sugita, Y.; Okamoto, Y. Replica-exchange molecular dynamics method for protein folding. *Chem. Phys. Lett.* **1999**, *314*, 141−151.
(6) Laio, A.; Parrinello, M. Escaping free-energy minima. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 12562−12566.
(7) Darve, E.; Rodríguez-Gómez, D.; Pohorille, A. Adaptive biasing force method for scalar and vector free energy calculations. *J. Chem. Phys.* **2008**, *128*, 144120.
(8) Torrie, G. M.; Valleau, J. P. Non-physical sampling distributions in Monte-Carlo free-energy estimation - umbrella sampling. *J. Comput. Phys.* **1977**, *23*, 187−199.
(9) Grubmüller, H.; Heymann, B.; Tavan, P. Ligand Binding: Molecular Mechanics Calculation of the Streptavidin-Biotin Rupture Force. *Science* **1996**, *271*, 997−999.
(10) Isralewitz, B.; Gao, M.; Schulten, K. Steered molecular dynamics and mechanical functions of proteins. *Curr. Opin. Struct. Biol.* **2001**, *11*, 224−230.
(11) Park, S.; Schulten, K. Calculating potentials of mean force from steered molecular dynamics simulations. *J. Chem. Phys.* **2004**, *120*, 5946−5961.
(12) Schlitter, J.; Engels, M.; Krüger, P.; Jacoby, E.; Wollmer, A. Targeted Molecular Dynamics Simulation of Conformational Change-Application to the T ↔ R Transition in Insulin. *Mol. Simul.* **1993**, *10*, 291−308.
(13) Schlitter, J.; Engels, M.; Krüger, P. Targeted Molecular Dynamics - A New Approach for Searching Pathways of Conformational Transitions. *J. Mol. Graphics* **1994**, *12*, 84−89.
(14) Schlitter, J.; Swegat, W.; Mülders, T. Distance-type reaction coordinates for modelling activated processes. *J. Mol. Model.* **2001**, *7*, 171−177.
(15) Berendsen, H. J. C. *Simulating the Physical World*; Cambridge University Press: Cambridge, 2007.
(16) Jarzynski, C. Nonequilibrium equality for free energy differences. *Phys. Rev. Lett.* **1997**, *78*, 2690−2693.
(17) Hendrix, D. A.; Jarzynski, C. A "fast growth" method of computing free energy differences. *J. Chem. Phys.* **2001**, *114*, 5974−5981.
(18) Hummer, G. Nonequilibrium Methods for Equilibrium Free Energy Calculations. In *Free Energy Calculations*, Chipot, C.; Pohorille, A., Eds; Springer: Berlin, 2007; 171−198
(19) Straub, J. E.; Borkovec, M.; Berne, B. J. Calculation of Dynamic Friction on Intramolecular Degrees of Freedom. *J. Phys. Chem.* **1987**, *91*, 4995−4998.

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182

(20) Straub, J. E.; Berne, B. J.; Roux, B. Spatial dependence of time-dependent friction for pair diffusion in a simple fluid. *J. Chem. Phys.* **1990**, *93*, 6804−6812.
(21) Koneshan, S.; Lynden-Bell, R. M.; Rasaiah, J. C. Friction coefficients of ions in aqueous solution at 25 degrees C. *J. Am. Chem. Soc.* **1998**, *120*, 12041−12050.
(22) Hummer, G. Position-dependent diffusion coefficients and free energies from Bayesian analysis of equilibrium and replica molecular dynamics simulations. *New J. Phys.* **2005**, *7*, 34.
(23) Morrone, J. A.; Li, J.; Berne, B. J. Interplay between Hydrodynamics and the Free Energy Surface in the Assembly of Nanoscale Hydrophobes. *J. Phys. Chem. B* **2012**, *116*, 378−389.
(24) Karim, O. A.; McCammon, J. A. Dynamics of a sodium chloride ion pair in water. *J. Am. Chem. Soc.* **1986**, *108*, 1762−1766.
(25) Ohtaki, H.; Radnai, T. Structure and dynamics of hydrated ions. *Chem. Rev.* **1993**, *93*, 1157−1204.
(26) Mullen, R. G.; Shea, J.-E.; Peters, B. Transmission Coefficients, Committors, and Solvent Coordinates in Ion-Pair Dissociation. *J. Chem. Theory Comput.* **2014**, *10*, 659−667.
(27) Zwanzig, R. *Nonequilibrium Statistical Mechanics*; Oxford University: Oxford, 2001.
(28) Meyer, H.; Voigtmann, T.; Schilling, T. On the non-stationary generalized Langevin equation. *J. Chem. Phys.* **2017**, *147*, 214110.
(29) Berry, M. V.; Robbins, J. M. Chaotic classical and half-classical adiabatic reactions: geometric magnetism and deterministic friction. *Proc. R. Soc. London, Ser. A* **1993**, *442*, 659−672.
(30) Jarzynski, C. Multiple-time-scale approach to ergodic adiabatic systems: Another look. *Phys. Rev. Lett.* **1993**, *71*, 839−842.
(31) Servantie, J.; Gaspard, P. Methods of calculation of a friction coefficient: application to nanotubes. *Phys. Rev. Lett.* **2003**, *91*, 185503.
(32) Following Zwanzig's derivation[27] of a Langevin equation in the case of a bilinearly coupled harmonic bath, one treats the restraint mode as an additional harmonic bath mode and obtains an analytic expression for the resulting "external" force f_c. Under the stiff spring approximation, the restraint can be modeled as a positional constraint.
(33) Mülders, T.; Krüger, P.; Swegat, W.; Schlitter, J. Free energy as the potential of mean constraint force. *J. Chem. Phys.* **1996**, *104*, 4869−4870.
(34) Daldrop, J. O.; Kowalik, B. G.; Netz, R. R. External Potential Modifies Friction of Molecular Solutes in Water. *Phys. Rev. X* **2017**, *7*, No. 041065.
(35) Pronk, S.; Páll, S.; Schulz, R.; Larsson, P.; Bjelkmar, P.; Apostolov, R.; Shirts, M. R.; Smith, J. C.; Kasson, P. M.; van der Spoel, D.; Hess, B.; Lindahl, E. GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics* **2013**, *29*, 845−854.
(36) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. Comparison of simple potential functions for simulating liquid water. *J. Chem. Phys.* **1983**, *79*, 926−935.
(37) Hornak, V.; Abel, R.; Okur, A.; Strockbine, B.; Roitberg, A.; Simmerling, C. Comparison of multiple Amber force fields and development of improved protein backbone parameters. *Proteins: Struct., Funct., Genet.* **2006**, *65*, 712−725.
(38) Miyamoto, S.; Kollman, P. A. Settle: An analytical version of the SHAKE and RATTLE algorithm for rigid water models. *J. Comput. Chem.* **1992**, *13*, 952−962.
(39) Darden, T.; York, D.; Pedersen, L. Particle mesh Ewald: An N log(N) method for Ewald sums in large systems. *J. Chem. Phys.* **1993**, *98*, 10089−10092.
(40) Bussi, G.; Donadio, D.; Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **2007**, *126*, No. 0141011.
(41) Parrinello, M.; Rahman, A. Polymorphic transitions in single crystals: A new molecular dynamics method. *J. Appl. Phys.* **1981**, *52*, 7182−7190.
(42) Humphrey, W.; Dalke, A.; Schulten, K. VMD − Visual Molecular Dynamics. *J. Mol. Graphics* **1996**, *14*, 33−38.
(43) Ryckaert, J. P.; Ciccotti, G.; Berendsen, H. J. C. Numerical-integration of cartesian equations of motions of a system with constraints-molecular dynamics of N-alkanes. *J. Comput. Phys.* **1977**, *23*, 327−341.
(44) Wolf, S.; Freier, E.; Gerwert, K. How Does a Membrane Protein Achieve a Vectorial Proton Transfer Via Water Molecules? *ChemPhysChem* **2008**, *9*, 2772−2778.
(45) Crooks, G. E. Nonequilibrium Measurements of Free Energy Differences for Microscopically Reversible Markovian Systems. *J. Stat. Phys.* **1998**, *90*, 1481−1487.
(46) Ytreberg, F. M.; Zuckerman, D. M. Efficient use of nonequilibrium measurement to estimate free energy differences for molecular systems. *J. Comput. Chem.* **2004**, *25*, 1749−1759.
(47) D'Agostino, R. B.; Belanger, A.; D'Agostino, R. B. A Suggestion for Using Powerful and Informative Tests of Normality. *Am. Stat.* **1990**, *44*, 316−321.
(48) Cellmer, T.; Henry, E. R.; Hofrichter, J.; Eaton, W. A. Measuring internal friction of an ultrafast-folding protein. *Proc. Natl. Acad. Sci. U. S. A.* **2008**, *105*, 18320−18325.
(49) Schulz, J. C. F.; Schmidt, L.; Best, R. B.; Dzubiella, J.; Netz, R. R. Peptide Chain Dynamics in Light and Heavy Water: Zooming in on Internal Friction. *J. Am. Chem. Soc.* **2012**, *134*, 6273−6279.
(50) Soranno, A.; Buchli, B.; Nettels, D.; Cheng, R. R.; Müller-Späth, S.; Pfeil, S. H.; Hoffmann, A.; Lipman, E. A.; Makarov, D. E.; Schuler, B. Quantifying internal friction in unfolded and intrinsically disordered proteins with single-molecule spectroscopy. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109*, 17800−17806.
(51) Erbaş, A.; Netz, R. R. Confinement-Dependent Friction in Peptide Bundles. *Biophys. J.* **2013**, *104*, 1285−1295.
(52) Echeverria, I.; Makarov, D. E.; Papoian, G. A. Concerted Dihedral Rotations Give Rise to Internal Friction in Unfolded Proteins. *J. Am. Chem. Soc.* **2014**, *136*, 8708−8713.
(53) Nakayama, T. THz frequency dynamics of network/guest atom systems: Liquid water, clathrates, and network glasses. *Nucl. Instrum. Methods Phys. Res., Sect. A* **2009**, *600*, 266−268.
(54) Kumar, S.; Rosenberg, J.; Bouzida, D.; Swendsen, R.; Kollman, P. The weighted histogram analysis method for free-energy calculations on biomolecules. I. The method. *J. Comput. Chem.* **1992**, *13*, 1011−1021.
(55) Sittel, F.; Stock, G. Perspective: Identification of Collective Coordinates and Metastable States of Protein Dynamics. *J. Chem. Phys.* **2018**, *149*, 150901.
(56) Ernst, M.; Wolf, S.; Stock, G. Identification and validation of reaction coordinates describing protein functional motion: Hierarchical dynamics of T4 Lysozyme. *J. Chem. Theory Comput.* **2017**, *13*, 5076−5088.
(57) Schaudinnus, N.; Lickert, B.; Biswas, M.; Stock, G. Global Langevin model of multidimensional biomolecular dynamics. *J. Chem. Phys.* **2016**, *145*, 184114.
(58) Bruce, N. J.; Ganotra, G. K.; Kokh, D. B.; Sadiq, S. K.; Wade, R. C. New approaches for computing ligand-receptor binding kinetics. *Curr. Opin. Struct. Biol.* **2018**, *49*, 1−10.
(59) Cheng, X.; McCoy, J. H.; Israelachvili, J. N.; Cohen, I. Imaging the Microscopic Structure of Shear Thinning and Thickening Colloidal Suspensions. *Science* **2011**, *333*, 1276−1279.

DOI: 10.1021/acs.jctc.8b00835
J. Chem. Theory Comput. 2018, 14, 6175−6182